

Lab 9: Instrumental Variables

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Study Overview: IGFBP-3 and Breast Tissue Density

Study Design

- **Population:** 139 premenopausal women
- **Main variables:**
 - Total dense breast tissue area (outcome)
 - Serum IGFBP-3 levels (exposure)
 - IGFBP3g genotype: AA, AC, CC (putative functional variant)

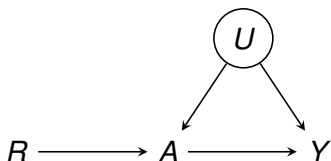
Motivation

- Examine relationship between IGFBP-3 (a regulator of IGF-1) and breast tissue characteristics

Concern

- Potential **confounding** by reproductive behaviors and other breast cancer risk factors

Instrumental Variable Setting



where

- Y = Total area of dense tissue in the breast (outcome),
- A = Serum IGFBP-3 levels (exposure),
- R = Putative functional gene for the protein IGFBP-3 (IV),
- U = ... (unobserved confounders)

IV Assumptions

- 1 **Relevance:** the instrumental variable R is associated with the exposure A .
- 2 **Exclusion Restriction:** the instrumental variable R affects the outcome Y *only through* the exposure A .
- 3 **Randomization:** the instrumental variable R is (conditionally) independent to the counterfactual outcome Y^a .

Question 1

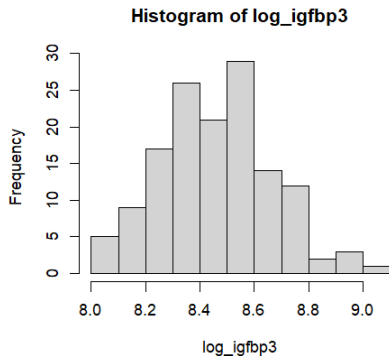
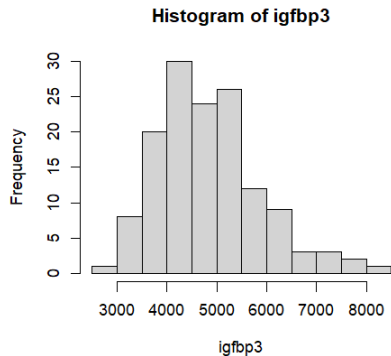
We want to test whether there is an **association** between the instrumental variable (`igfbp3g`) and the outcome (`lucent`). Under the usual IV assumptions, any such association should occur *only through* the exposure (`igfbp3`).

```
lm_model <- lm(log_lucent ~ as.numeric(igfbp3g), data = data)
```

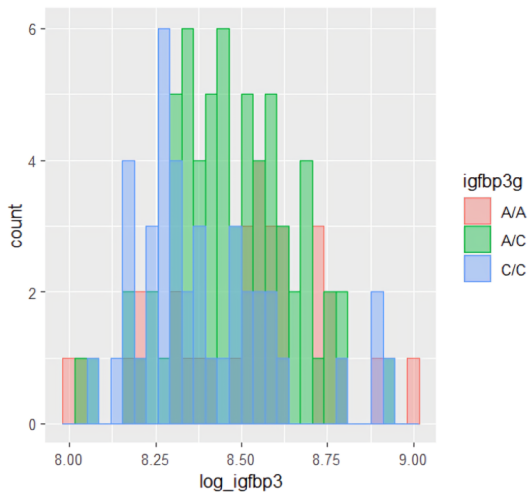
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.43102	0.20365	21.758	<2e-16 ***
<code>as.numeric(igfbp3g)</code>	-0.15431	0.06468	-2.386	0.0184 *

Question 2 - log IGFBP3 histogram



Question 2 - log IGFBP3 histogram by genotype



Question 2 - Testing Relevance

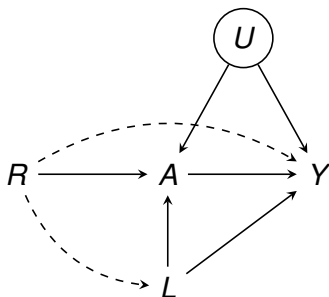
In the traditional 2SLS we usually need that the coefficient of the regression of the exposure onto the IV is statistically significant.

```
lm_model2 <- lm(log_igfbp3 ~ as.numeric(igfbp3g), data = data)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	8.65484	0.07484	115.643	< 2e-16	***
as.numeric(igfbp3g)	-0.06332	0.02377	-2.664	0.00865	**

Question 3 - DAG



Variables:

- R : IGFBP3g genotype (instrument)
- A : log(serum IGFBP-3) (exposure)
- Y : log(lucent area) (outcome)
- U : unobserved confounders
- L : age, afbcat, brfcat, parcat, ev_pil (observed confounders)

Question 3 - DAG and Assumptions

- **Relevance (Instrument affects the exposure):**

$$R \not\perp A \mid L$$

The genotype must be associated with serum IGFBP-3 levels after adjusting for observed covariates.

- **Exclusion Restriction (Instrument affects the outcome only through the exposure):**

$$R \perp Y \mid A, L, U$$

Genotype should not affect lucent area through any path except via serum IGFBP-3.

This is **potentially violated** if, for example, genotype affects fertility traits like parity or age at first birth, which in turn affect lucent area.

Question 3 - DAG and Assumptions Cont'd

- **Independence (Instrument is independent of unobserved confounders):**

$$R \perp\!\!\!\perp U \mid L$$

The genotype is randomized by Mendelian inheritance (i.e., no confounding between genotype and unmeasured factors that influence both the exposure and the outcome).

Question 4 - IV Analysis

Estimand

$$\psi = \frac{E\{\text{Cov}(R, Y | L)\}}{E\{\text{Cov}(R, A | L)\}}$$

Efficient Influence Function

$$\frac{\{R - E(R | L)\}[Y - E(Y | L) - \psi\{A - E(A | L)\}]}{E[\{R - E(R | L)\}\{A - E(A | L)\}]}$$

A de-biased estimator which sets its sample average to zero equals:

$$\hat{\psi} = \frac{\sum_{i=1}^n \{R_i - \hat{E}(R_i | L_i)\} \{Y_i - \hat{E}(Y_i | L_i)\}}{\sum_{i=1}^n \{R_i - \hat{E}(R_i | L_i)\} \{A_i - \hat{E}(A_i | L_i)\}}$$

Question 4 - IV Analysis

Method	Estimate	CI_Lower	CI_Upper
<chr>	<chr>	<chr>	<chr>
1 Eff. Influence Function	1.261	1.001	1.589

Interpretation

A 10% increase in level of serum IGFBP-3 increases the geometric mean of the lucent area by 26.1%

(= $\exp(2.44 * \log(1.1))$) [95% CI: 0.1% to 58.9%].

Question 5 - IV Analysis with confounders

Method	Estimate	CI_Lower	CI_Upper
<chr>	<chr>	<chr>	<chr>
1 Eff. Influence Function	1.249	1.034	1.509

Interpretation

A 10% increase in level of serum IGFBP-3 increases the geometric mean of the lucent area by 24.9%

(= $\exp(2.33 * \log(1.1))$) [95% CI: 3.4% to 50.9%].

Question 6 - Assumption Lean Regression

Method	Estimate	CI_Lower	CI_Upper
<chr>	<chr>	<chr>	<chr>
1 Linear Model	1.045	0.998	1.094
2 Eff. Influence Function	1.045	0.998	1.095

Interpretation

A 10% increase in level of serum IGFBP-3 increases the geometric mean of the lucent area by 4.45%

(= $\exp(0.46 * \log(1.1))$) [95% CI: -0.2% to 9.27%].

APPENDIX

From Model to Estimand

Consider model

$$E(Y | A, L, R, U) = \psi A + g(L, U)$$

for some unknown function $g(\cdot)$.

$$\begin{aligned}\text{Cov}(R, Y | L) &= E[RY | L] - E[R | L]E[Y | L] \\&= E[RE[Y | A, L, R, U] | L] - E[R | L]E[E[Y | A, L, R, U] | L] \\&= E[R(\psi A + g(L, U)) | L] - E[R | L]E[(\psi A + g(L, U)) | L] \\&= \psi E[RA | L] + E[Rg(L, U) | L] \\&\quad - E[R | L] \left(\psi E[A | L] + E[g(L, U) | L] \right) \\&= \psi E[RA | L] + E[R | L]g(L, U) \\&\quad - \psi E[R | L]E[A | L] - E[R | L]g(L, U) \\&= \psi \text{Cov}(R, A | L)\end{aligned}$$

From Model to Estimand - Cont'd

$$\psi = \frac{\text{Cov}(R, Y \mid L)}{\text{Cov}(R, A \mid L)}$$

But since by the modeling assumption ψ is constant, we average across L :

$$E[\text{Cov}(R, Y \mid L)] = E[\psi \text{Cov}(R, A \mid L)]$$

$$\psi = \frac{E\{\text{Cov}(R, Y \mid L)\}}{E\{\text{Cov}(R, A \mid L)\}}$$

When Model is Misspecified

When

$$E(Y | A, L, R, U) = \psi(L)A + g(L, U)$$

then

$$\psi = \frac{E\{\text{Cov}(R, Y | L)\}}{E\{\text{Cov}(R, A | L)\}} = \frac{E\{w(L)\psi(L)\}}{E\{w(L)\}}$$

reduces to a weighted average of exposure effects, for weights $w(L) = \text{Cov}(R, A | L)$.

Following the same derivation as before we arrive at:

$$\text{Cov}(R, Y | L) = \psi(L) \text{Cov}(R, A | L)$$

When Model is Misspecified

Take expectations over L on both sides:

$$E[\text{Cov}(R, Y | L)] = E[\psi(L) \text{Cov}(R, A | L)].$$

So we can write:

$$E[\text{Cov}(R, Y | L)] = E[\psi(L)w(L)],$$

and we already know:

$$E[\text{Cov}(R, A | L)] = E[w(L)].$$

Thus:

$$\psi = \frac{E[\text{Cov}(R, Y | L)]}{E[\text{Cov}(R, A | L)]} = \frac{E[\psi(L)w(L)]}{E[w(L)]}.$$

Interpretation and Geometric Mean

For a variable X , the geometric mean is defined as:

$$\text{GM}(X) = \left(\prod_{i=1}^n X_i \right)^{1/n} = \exp \left(\frac{1}{n} \sum_{i=1}^n \log X_i \right)$$

That is, the exponential of the average of the log-transformed values.

Consider the log-log model

$$\log(Y) = \beta_0 + \beta_1 \log(X) + \epsilon.$$

Interpretation and Geometric Mean

The expected value of $\log(Y)$ is:

$$\underbrace{\mathbb{E}[\log(Y)]}_{\log(GM(Y))} = \beta_0 + \beta_1 \cdot \underbrace{\mathbb{E}[\log(X)]}_{\log(GM(X))}$$

Therefore we can write:

$$\begin{aligned} GM(Y) &= \exp(\beta_0 + \beta_1 \cdot \log(GM(X))) \\ &= \exp(\beta_0) \cdot GM(X)^{\beta_1} \end{aligned}$$

Interpretation and Geometric Mean

Geometric Mean of $Y \mid X$

$$\text{GM}(Y \mid X) = \exp(\mathbb{E}[\log Y \mid X]) = \exp(\beta_0 + \beta_1 \log X) = \exp(\beta_0) X^{\beta_1}.$$

Define the geometric mean under a 10% increase in X

Now, suppose $X \rightarrow X \cdot 1.1$, i.e., a 10% increase in X .

$$\text{GM}(Y \mid X \cdot 1.1) = \exp(\beta_0) (1.1X)^{\beta_1} = \exp(\beta_0) X^{\beta_1} \cdot (1.1)^{\beta_1}$$

So the relative change in the geometric mean is:

$$\frac{\text{GM}(Y \mid 1.1 \cdot X)}{\text{GM}(Y \mid X)} = \frac{\exp(\beta_0) X^{\beta_1} \cdot (1.1)^{\beta_1}}{\exp(\beta_0) X^{\beta_1}} = (1.1)^{\beta_1} = \exp(\beta_1 \cdot \log(1.1))$$